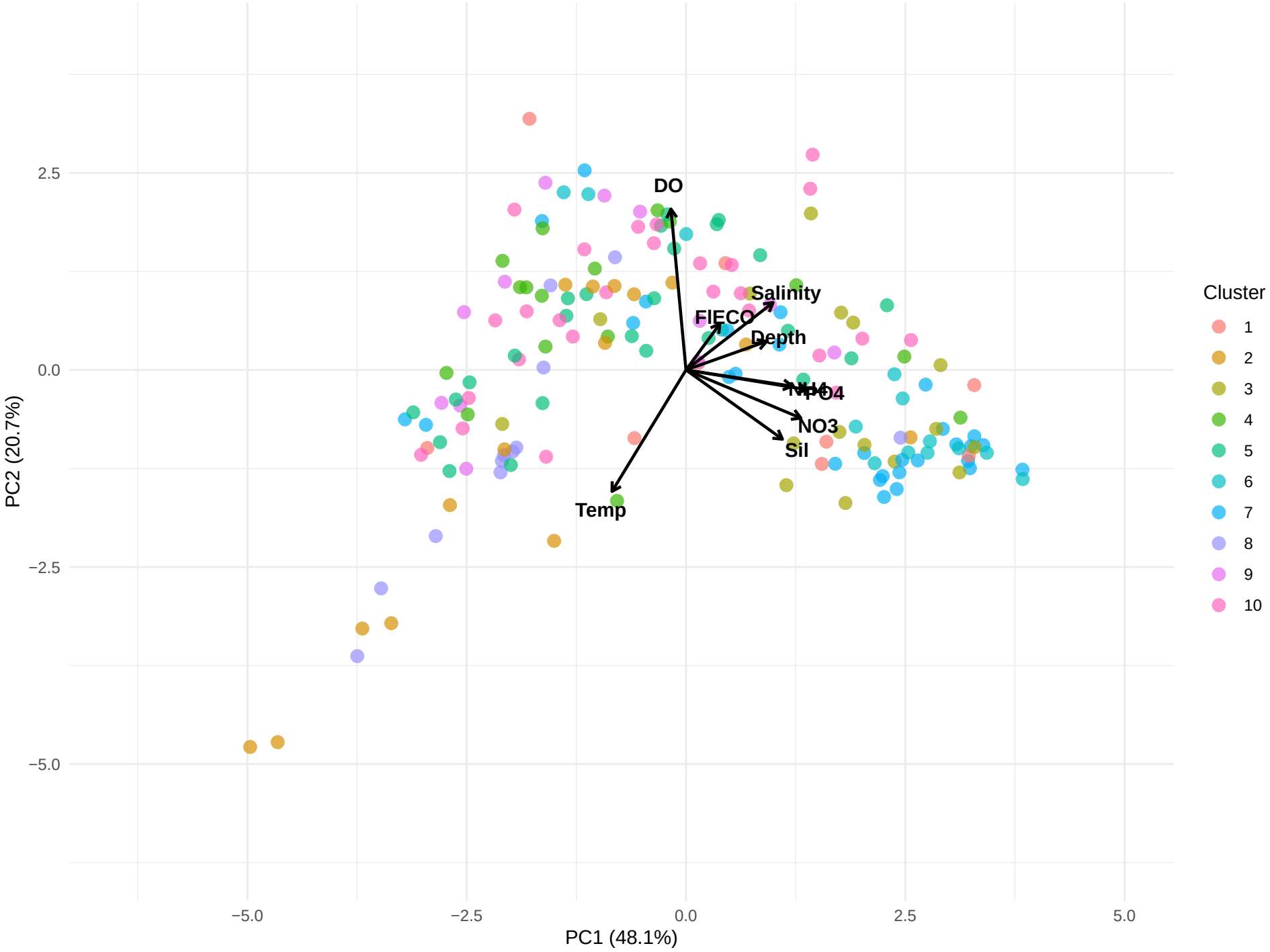
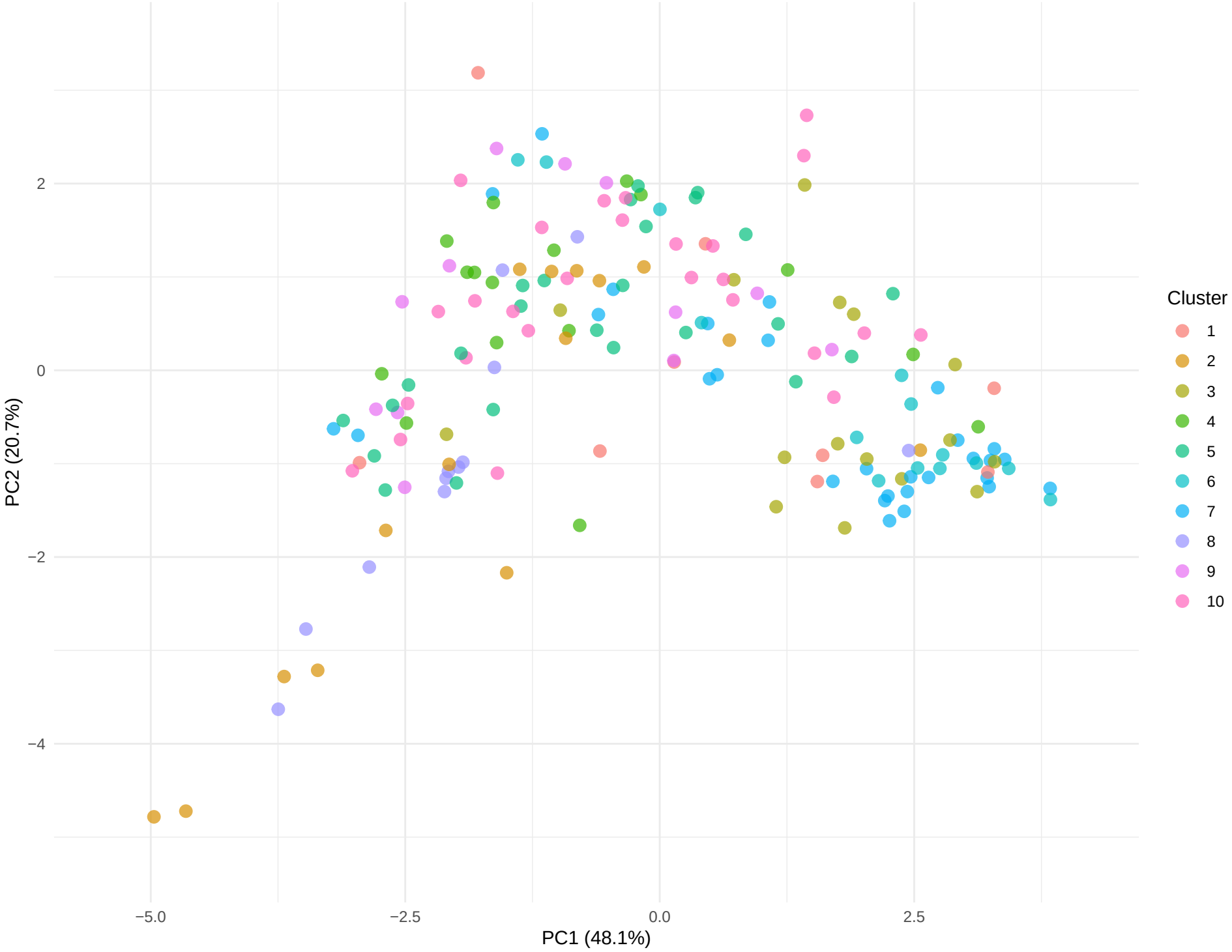
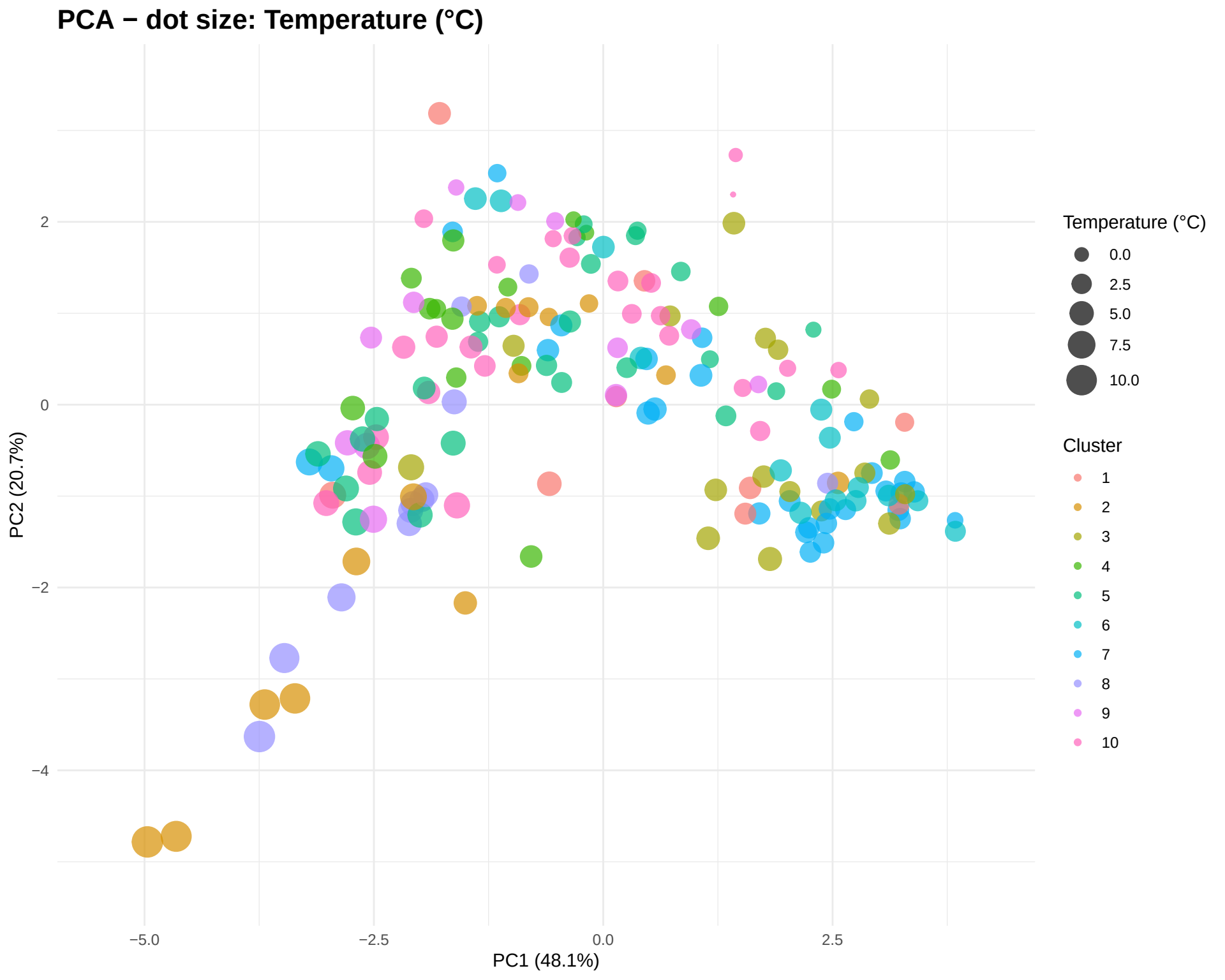
PCA of Environmental Variables (10 clusters)

log1p−transformed nutrients + scaled; 9 variables

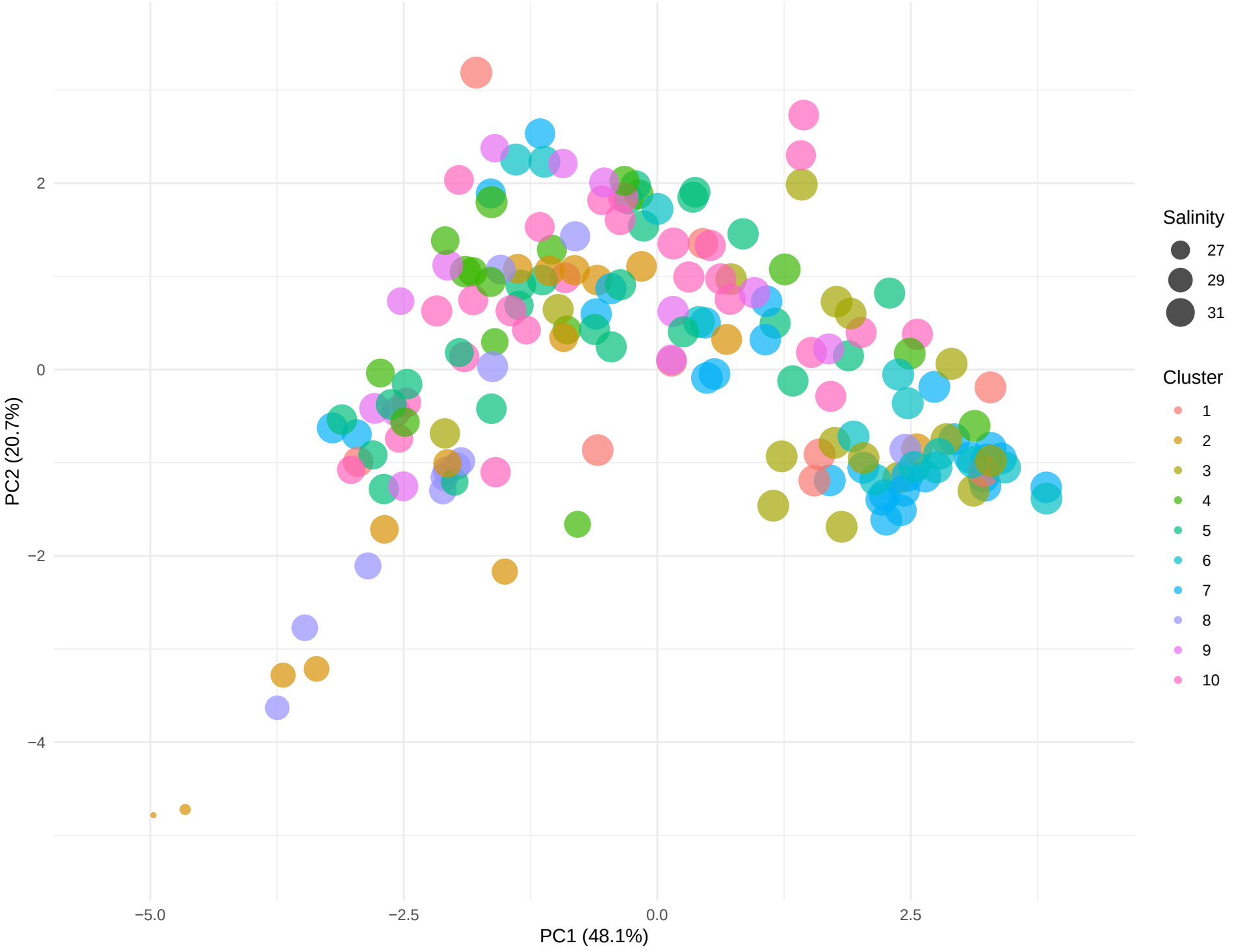


PCA of Environmental Variables (points only)

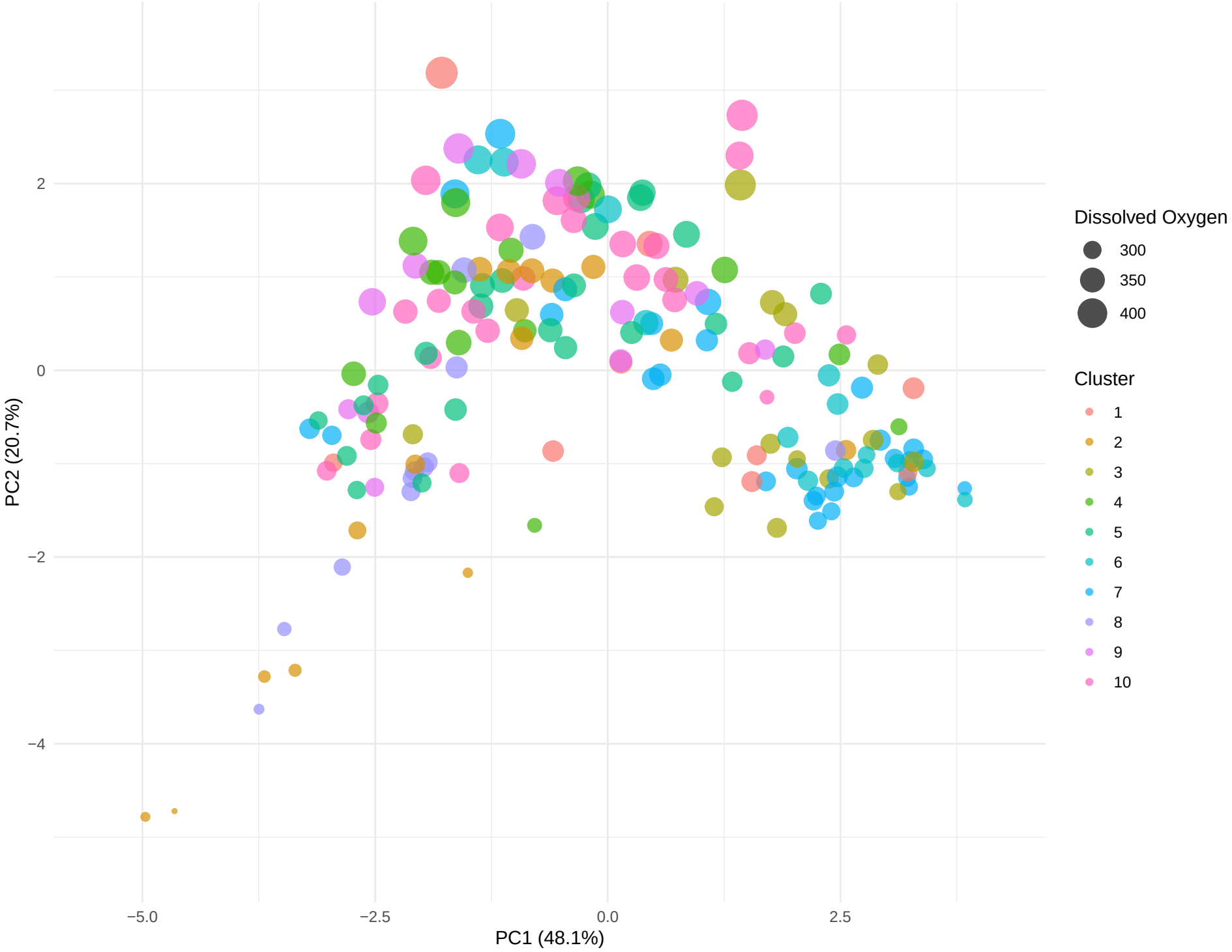


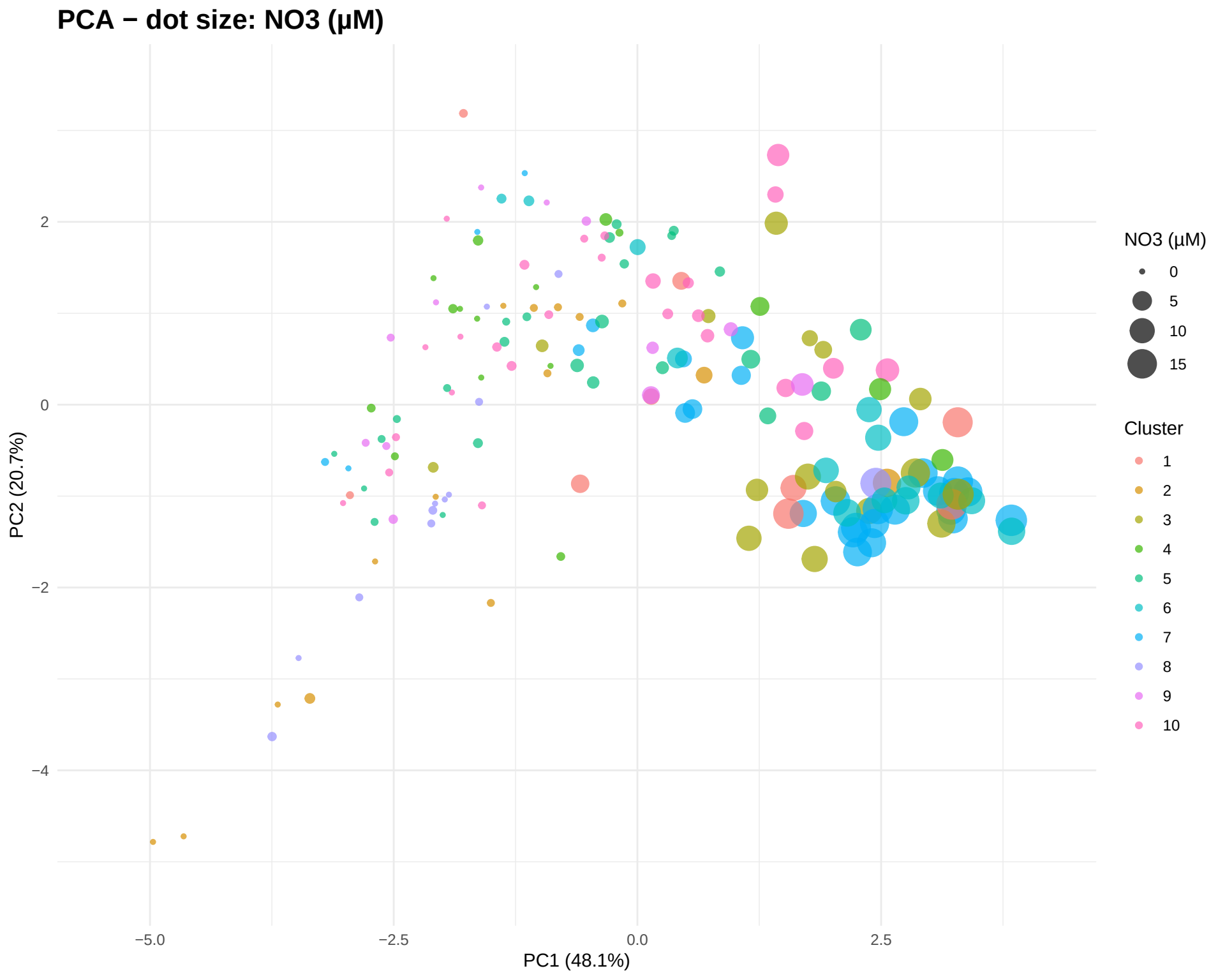

PCA − dot size: Temperature (°C)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Temperature (°C)
0.0
2.5
5.0
7.5
10.0
Cluster
1
2
3
4
5
6
7
8
9
10

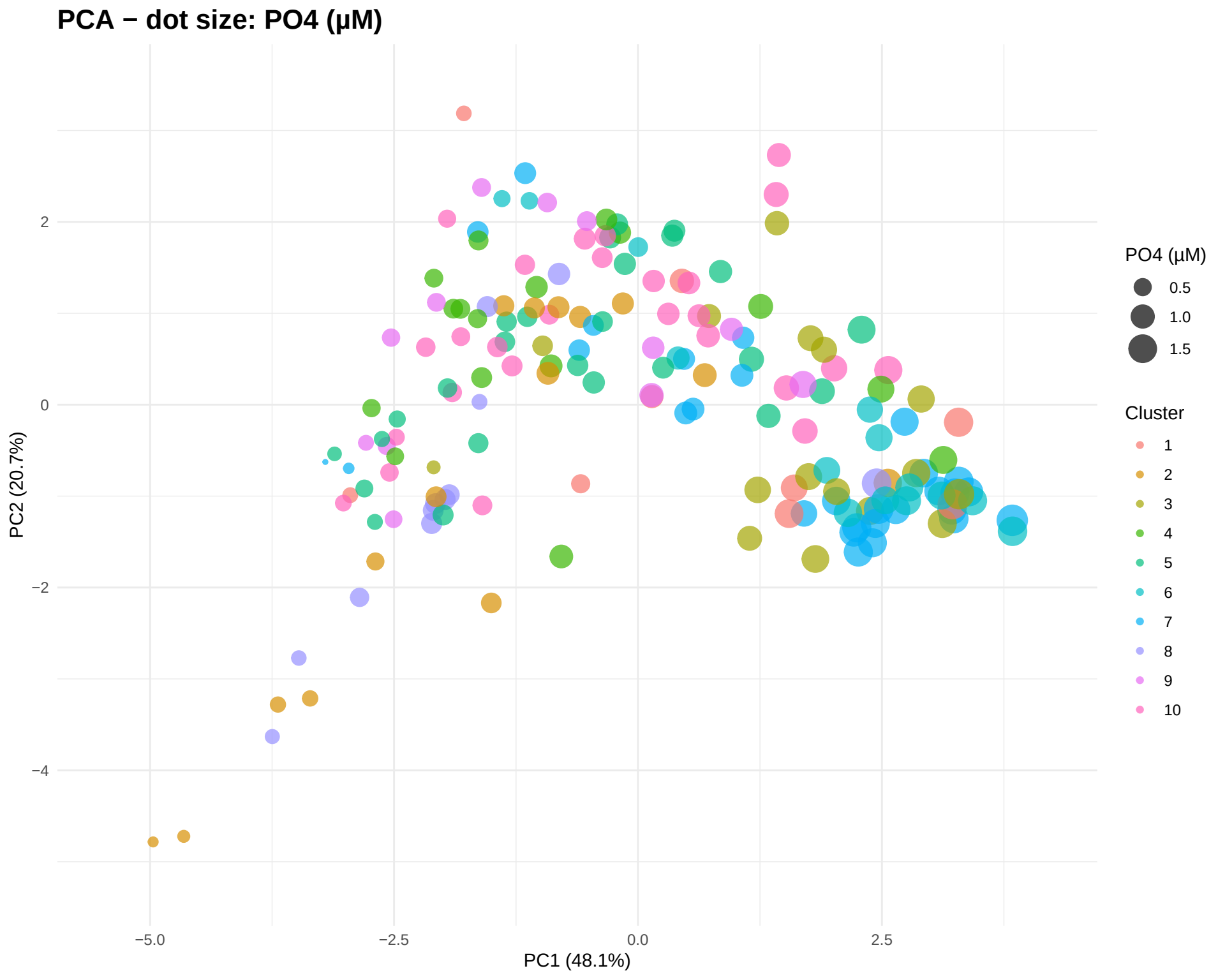
PCA − dot size: Salinity

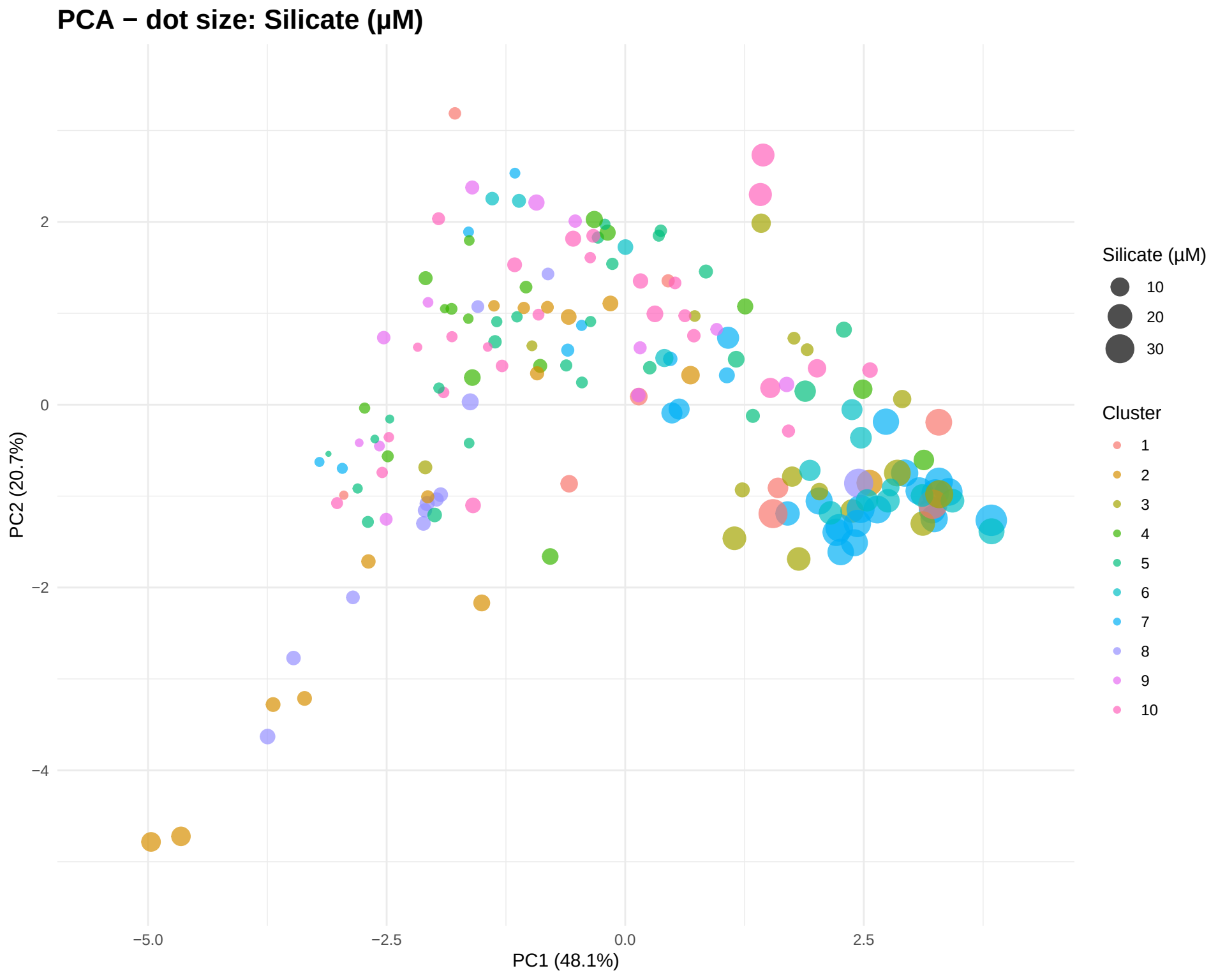


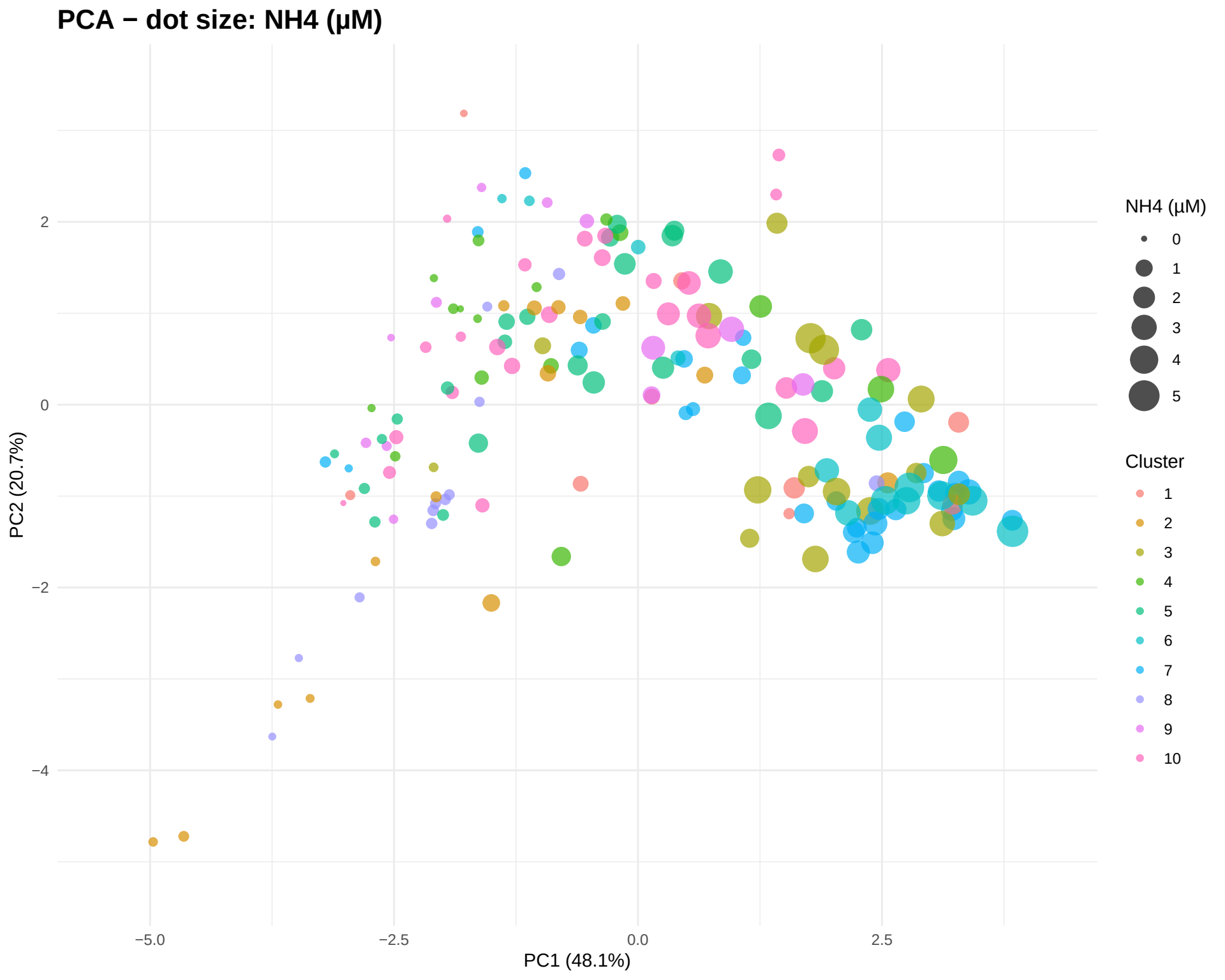
PCA − dot size: Dissolved Oxygen

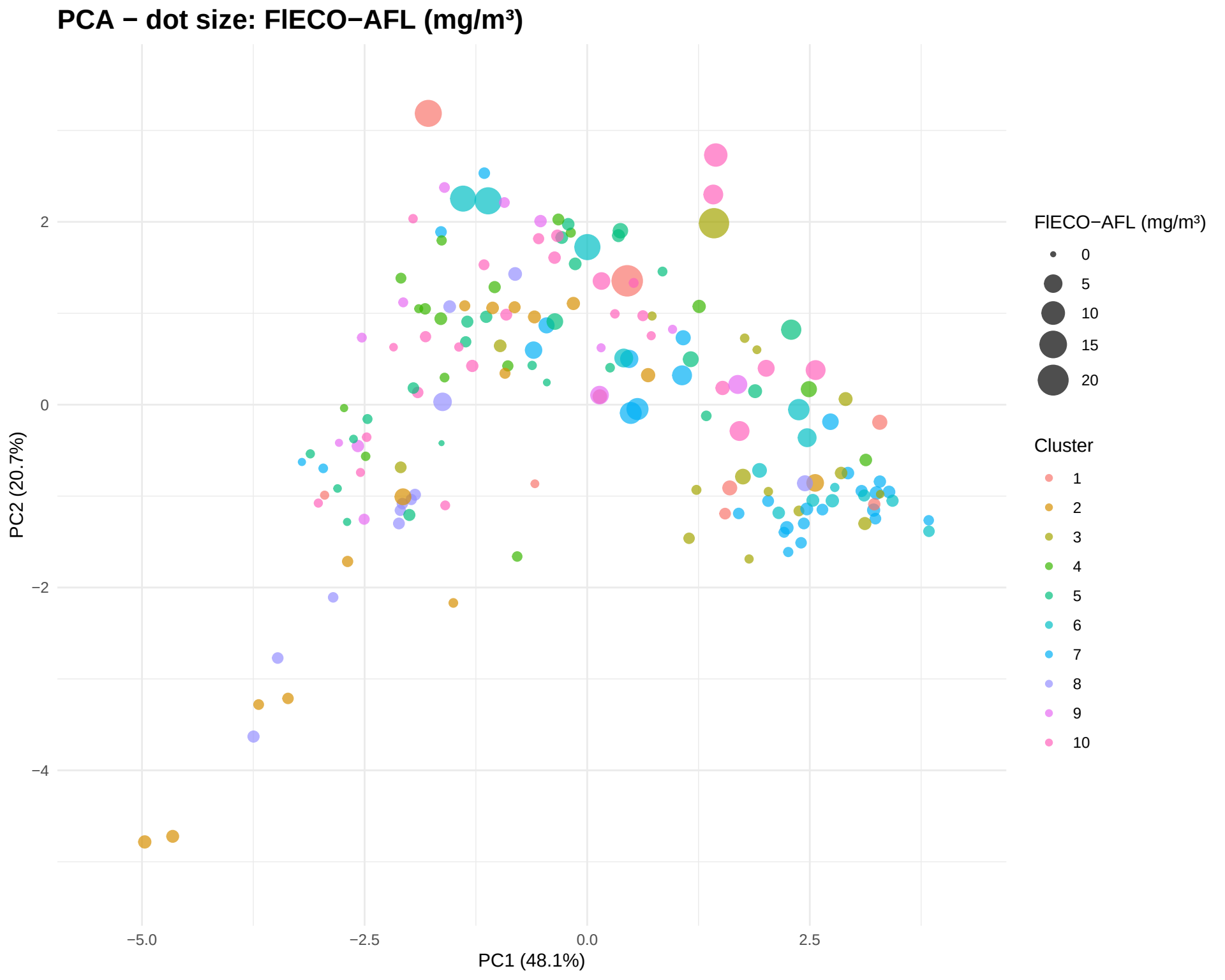


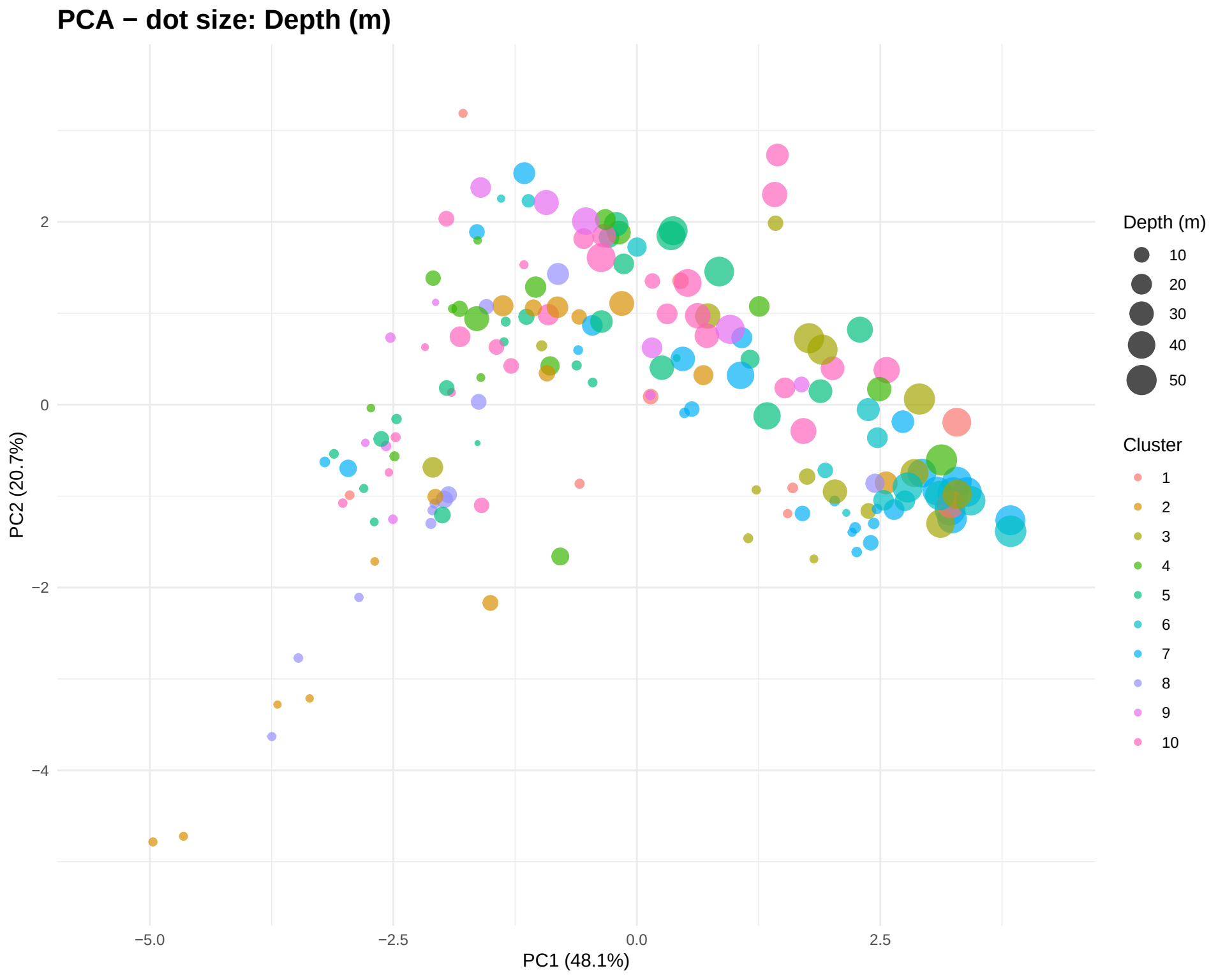

PCA − dot size: NO3 (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
NO3 (µM)
0
5
10
15
Cluster
1
2
3
4
5
6
7
8
9
10

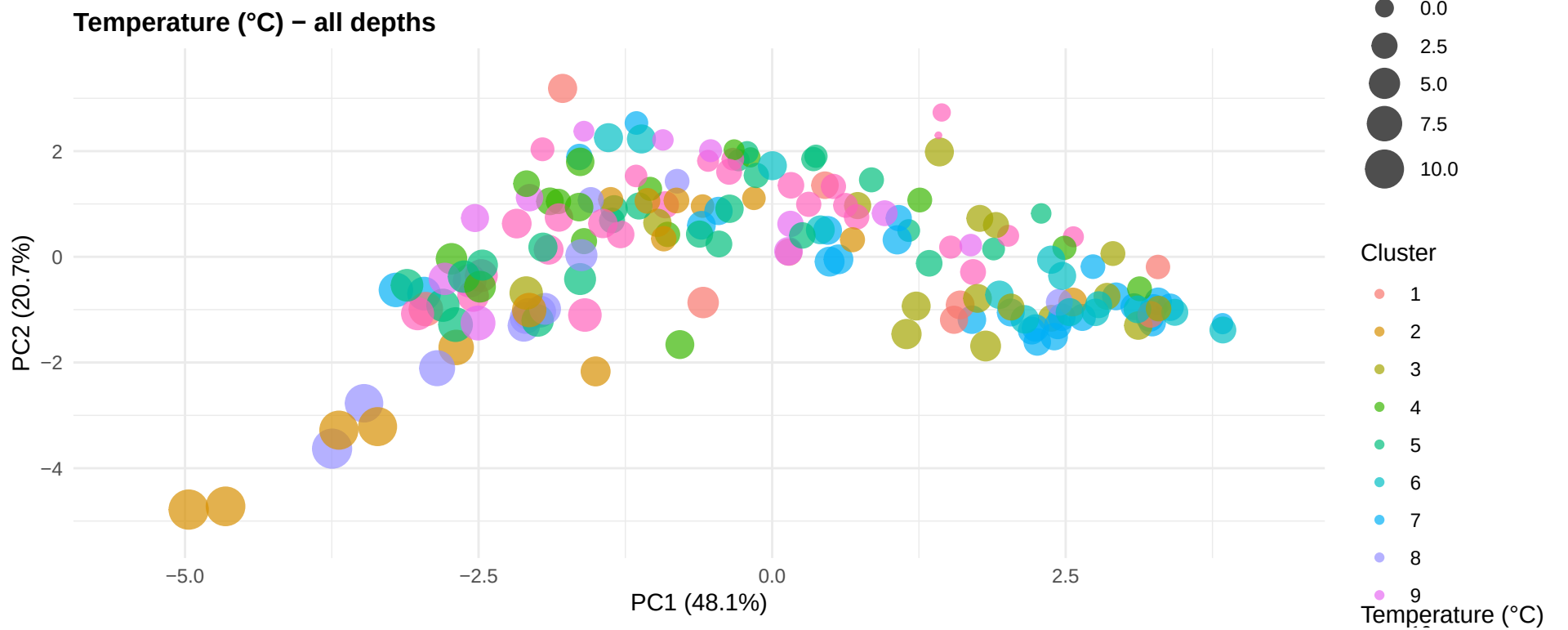

PCA − dot size: PO4 (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
PO4 (µM)
0.5
1.0
1.5
Cluster
1
2
3
4
5
6
7
8
9
10

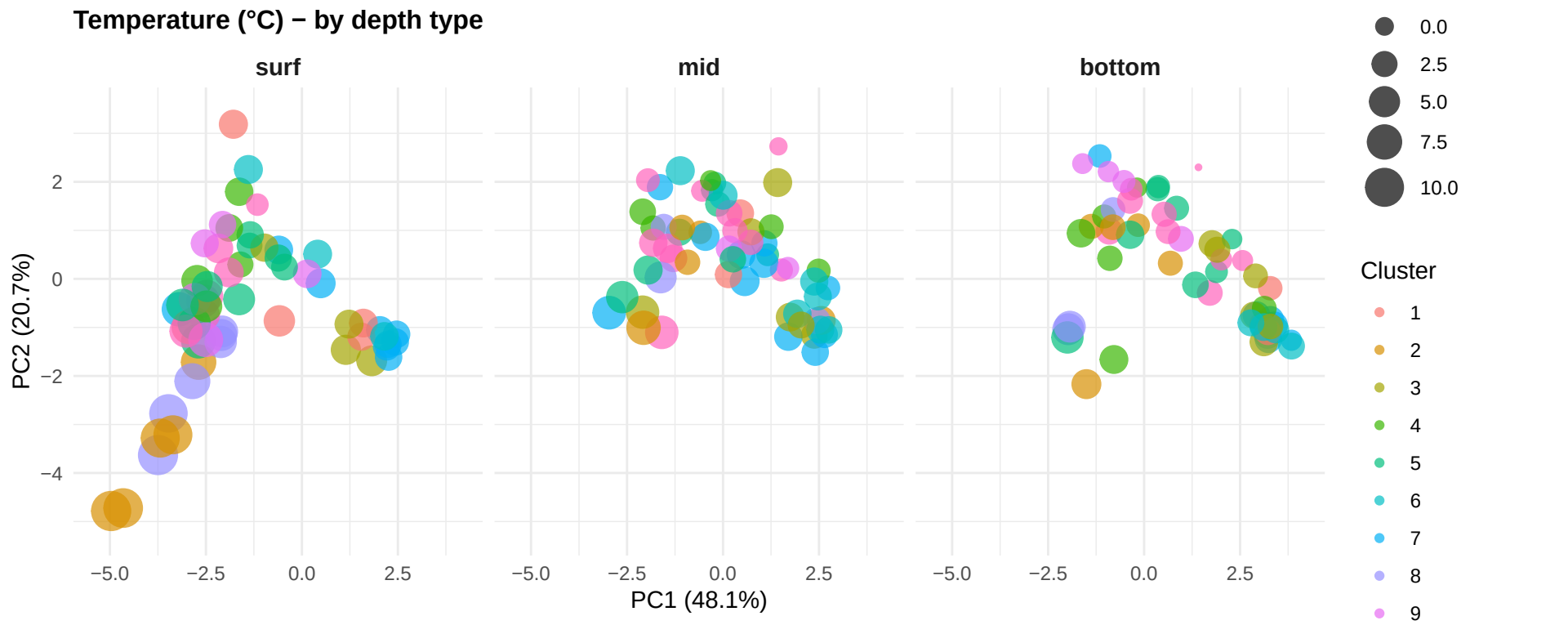

PCA − dot size: Silicate (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Silicate (µM)
10
20
30
Cluster
1
2
3
4
5
6
7
8
9
10

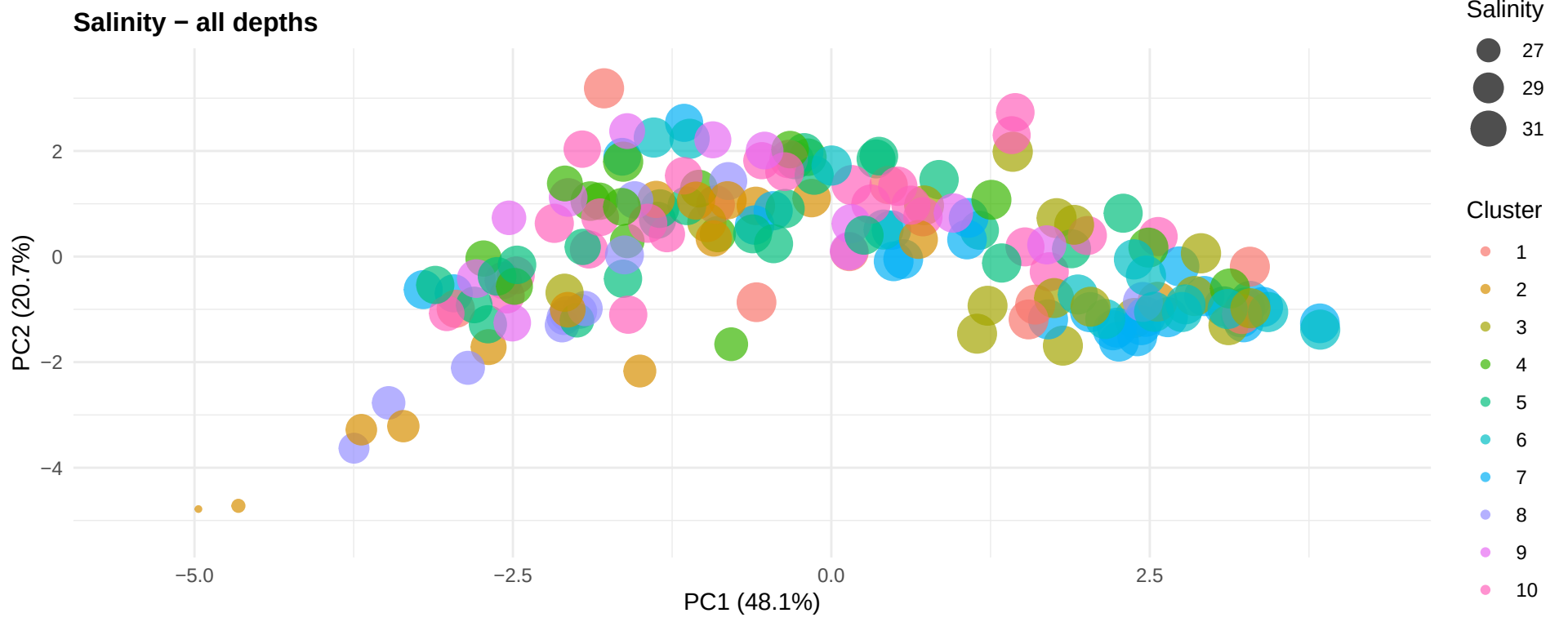

PCA − dot size: NH4 (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
NH4 (µM)
0
1
2
3
4
5
Cluster
1
2
3
4
5
6
7
8
9
10

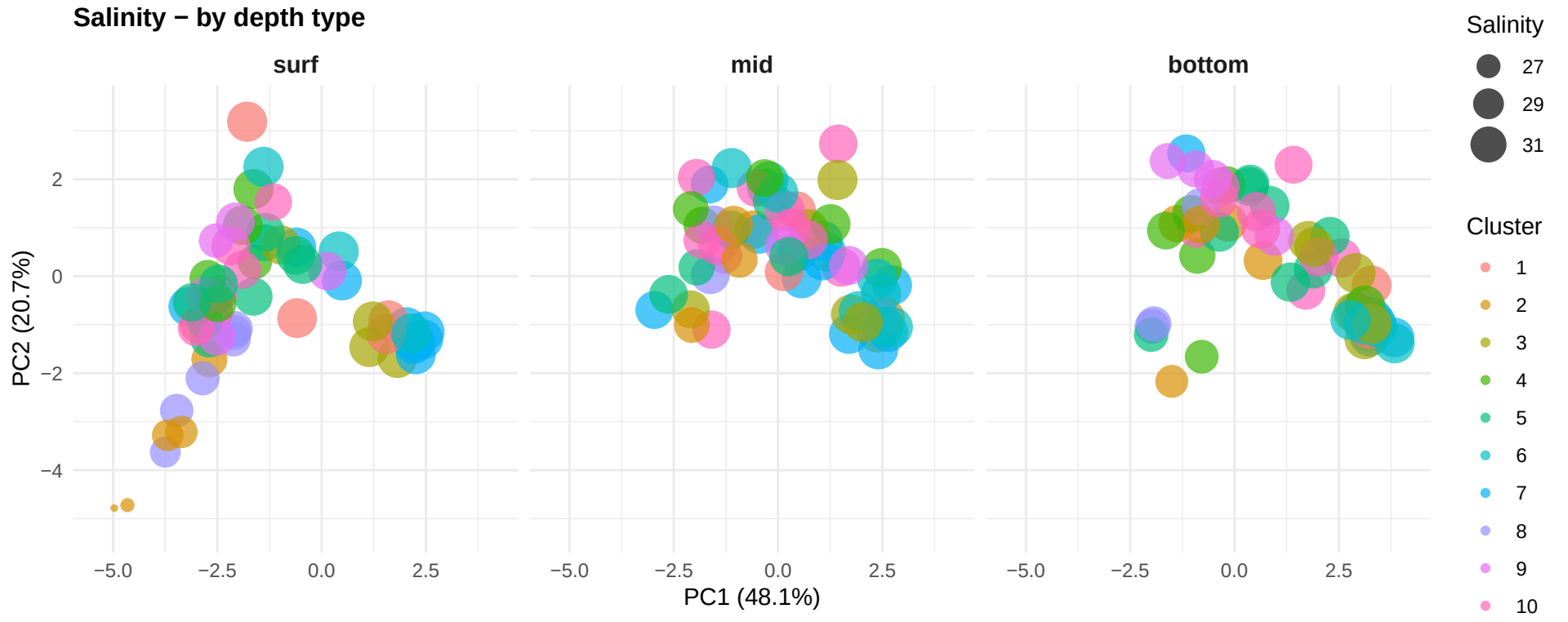

PCA − dot size: FlECO−AFL (mg/m³)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
FlECO−AFL (mg/m³)
0
5
10
15
20
Cluster
1
2
3
4
5
6
7
8
9
10

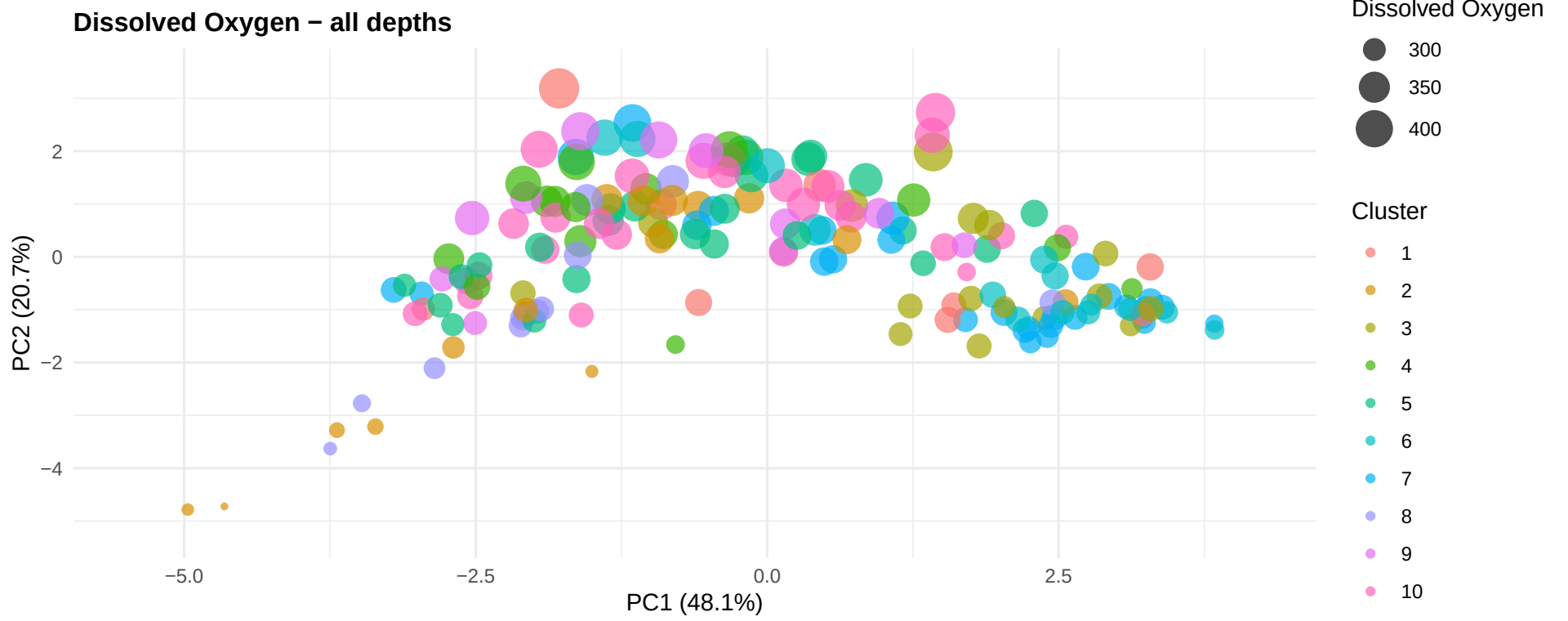

PCA − dot size: Depth (m)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Depth (m)
10
20
30
40
50
Cluster
1
2
3
4
5
6
7
8
9
10

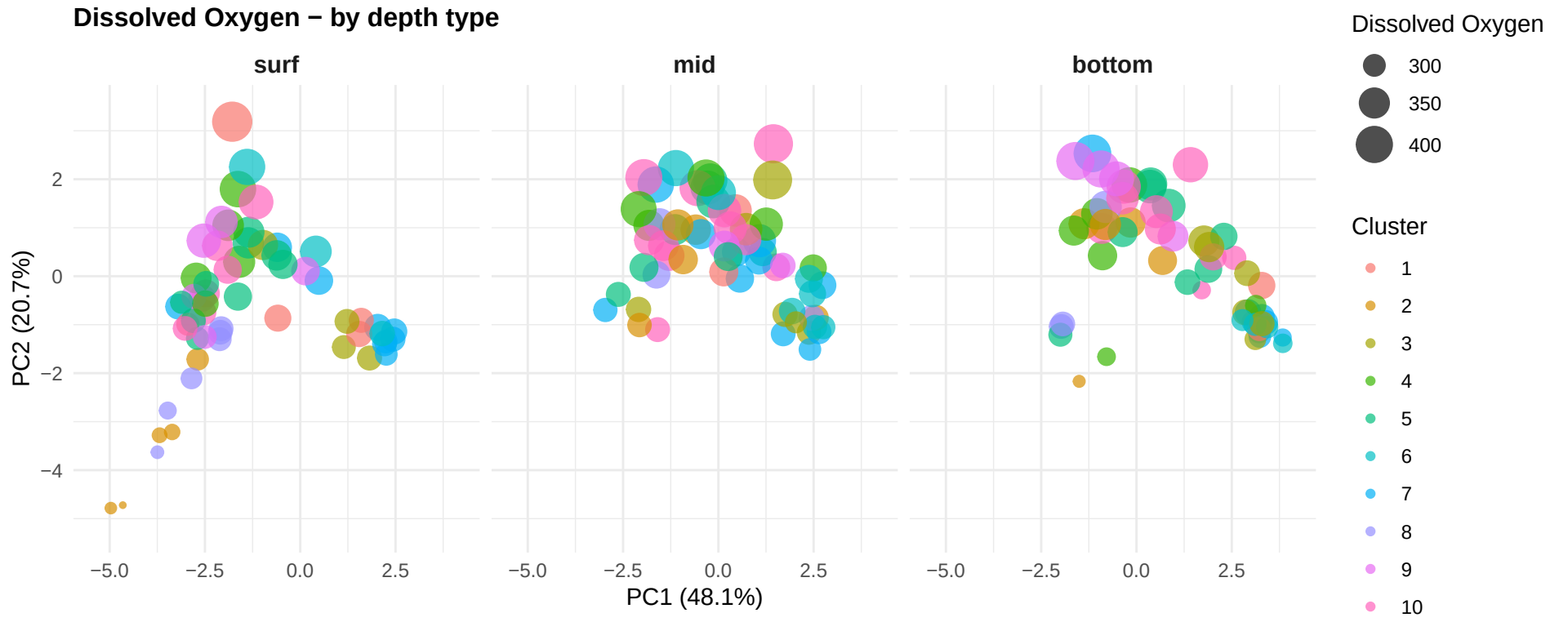

Temperature (°C) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Temperature (°C)
0.0
2.5
5.0
7.5
10.0
Cluster
1
2
3
4
5
6
7
8
9

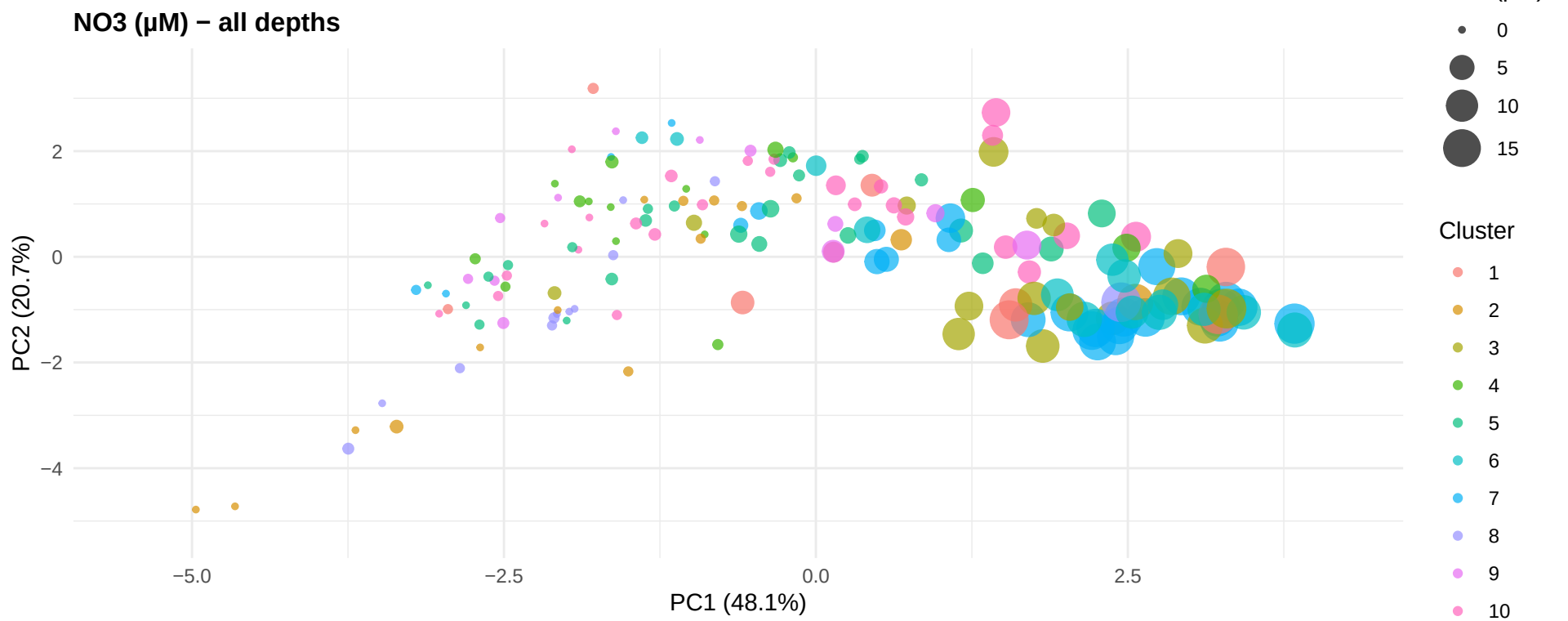

Temperature (°C) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Temperature (°C)
0.0
2.5
5.0
7.5
10.0
Cluster
1
2
3
4
5
6
7
8
9

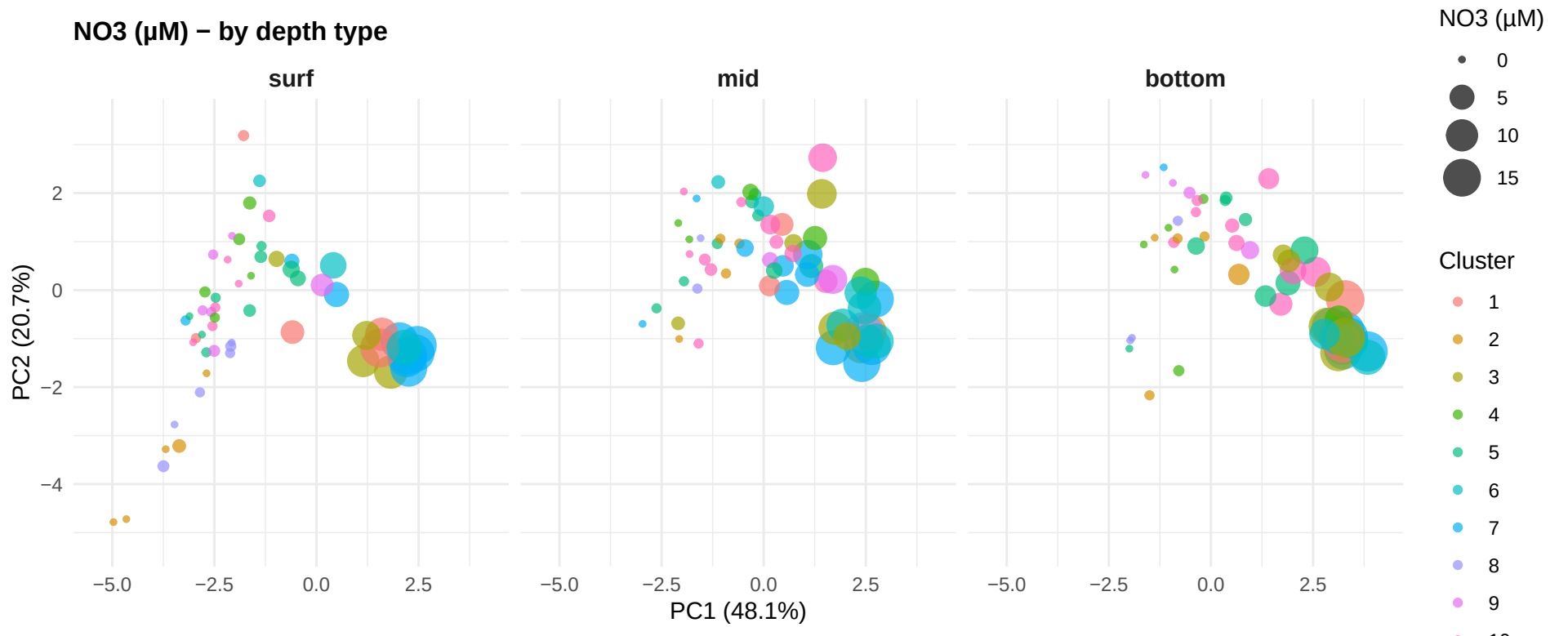

Salinity − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Salinity
27
29
31
Cluster
1
2
3
4
5
6
7
8
9
10

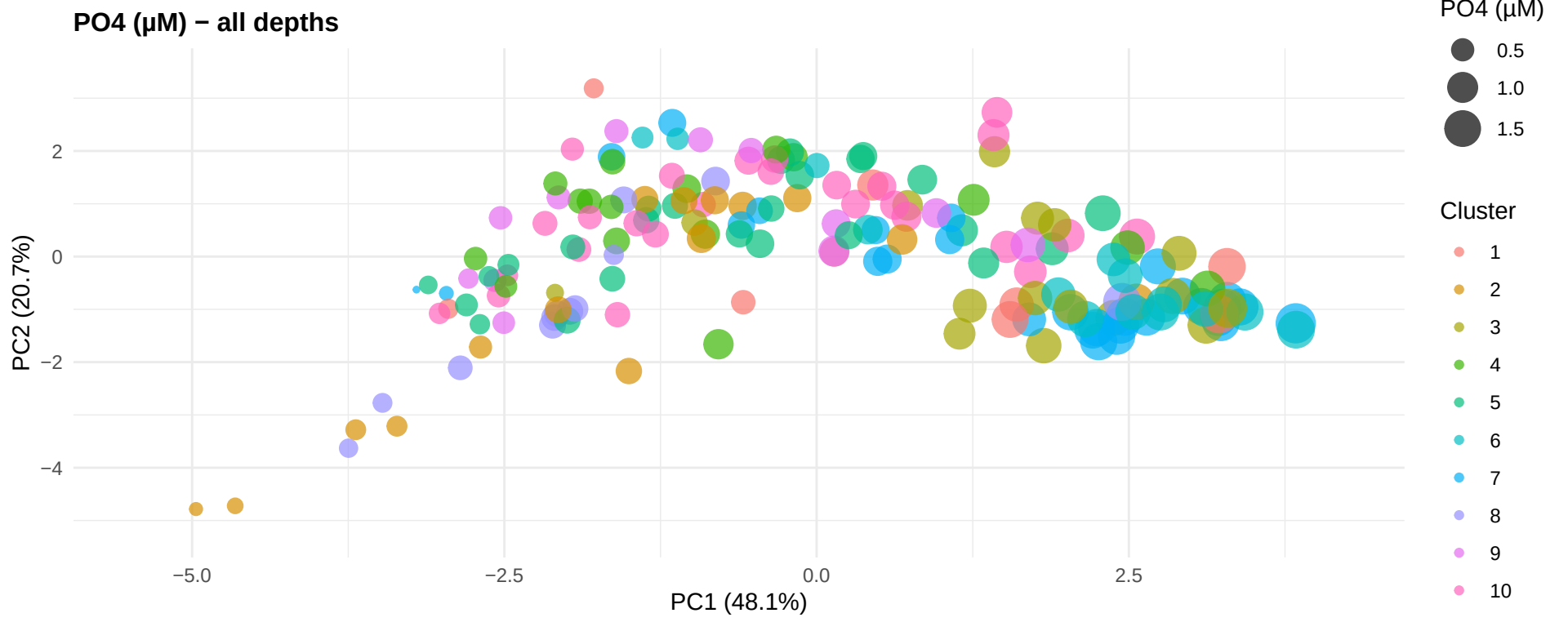

Salinity − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Salinity
27
29
31
Cluster
1
2
3
4
5
6
7
8
9
10

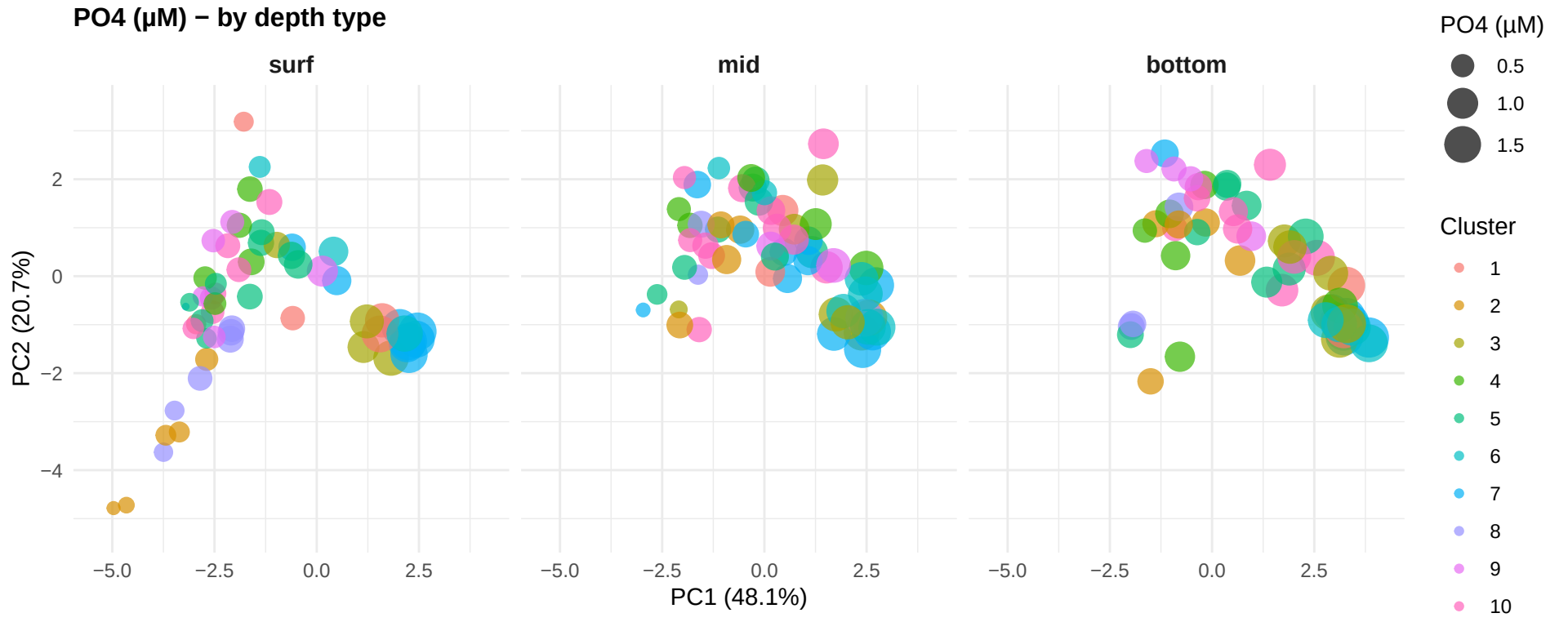

Dissolved Oxygen − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Dissolved Oxygen
300
350
400
Cluster
1
2
3
4
5
6
7
8
9
10

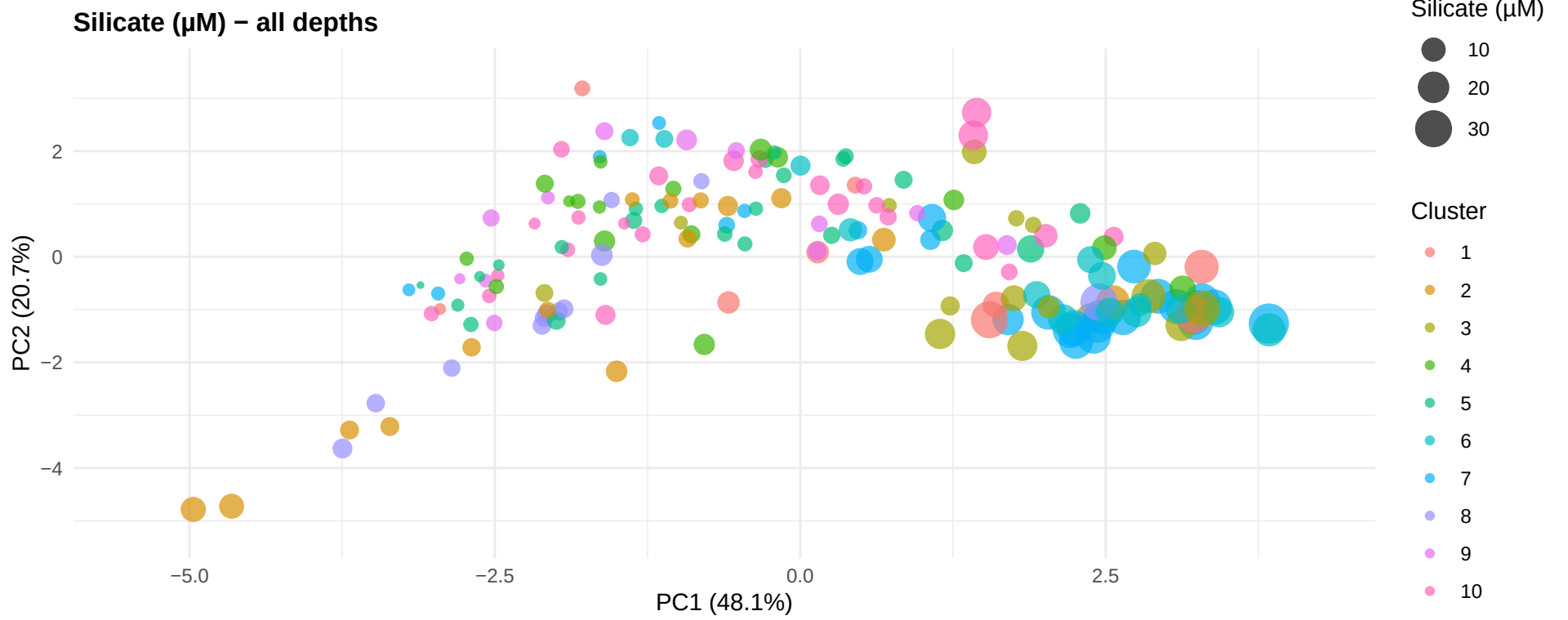

Dissolved Oxygen − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Dissolved Oxygen
300
350
400
Cluster
1
2
3
4
5
6
7
8
9
10

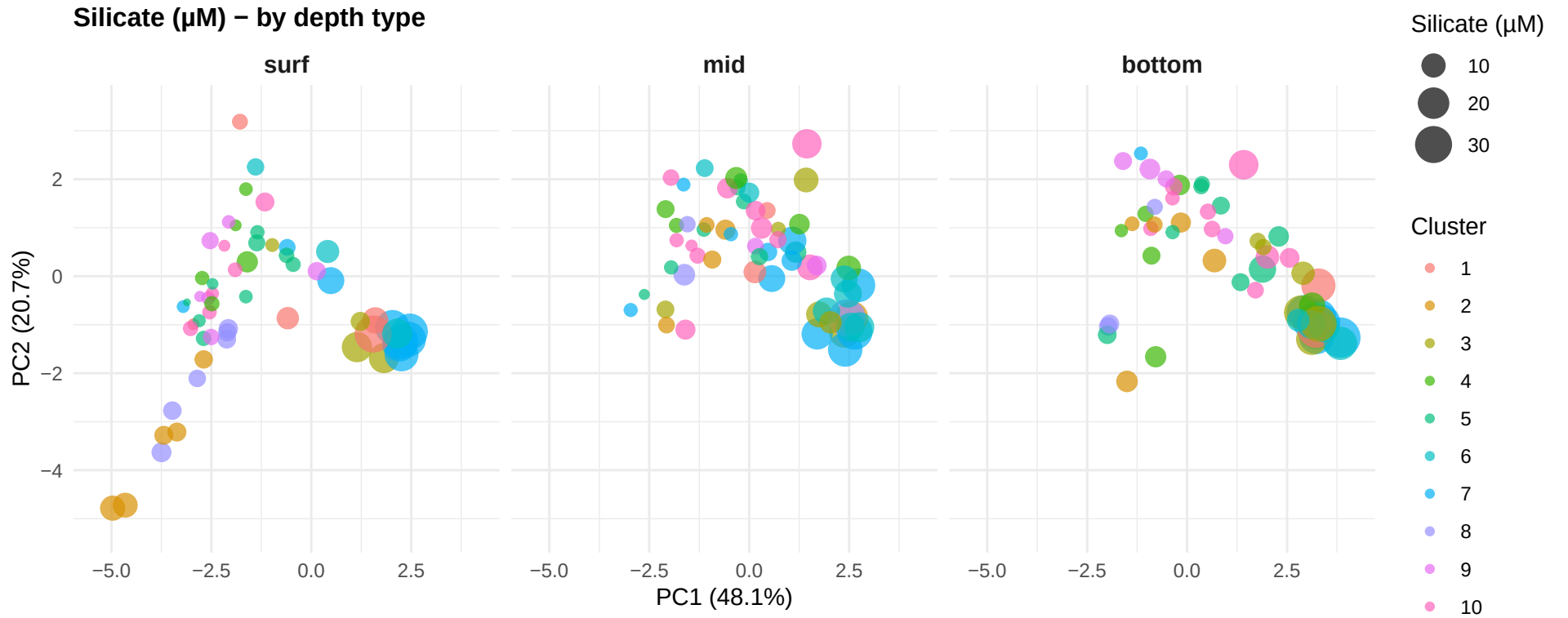

NO3 (µM) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
0
5
10
15
Cluster
1
2
3
4
5
6
7
8
9
10

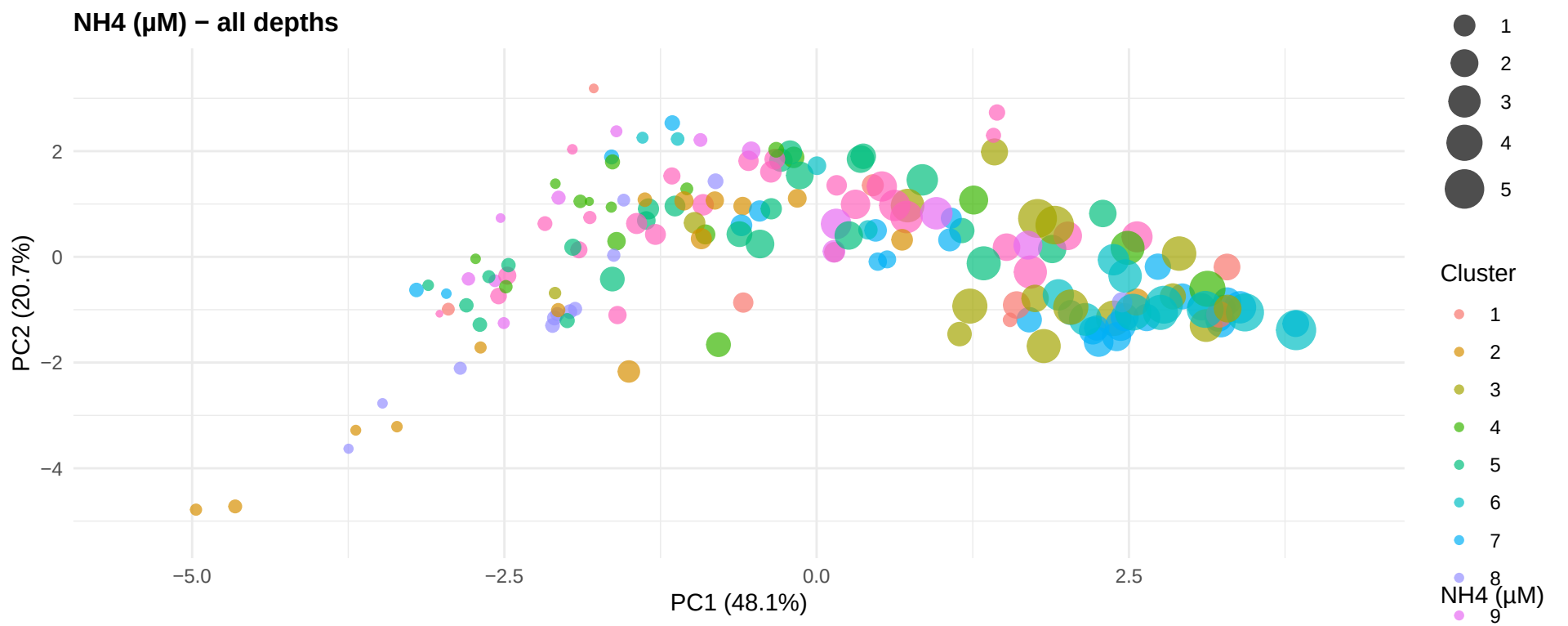

NO3 (µM) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
NO3 (µM)
0
5
10
15
Cluster
1
2
3
4
5
6
7
8
9

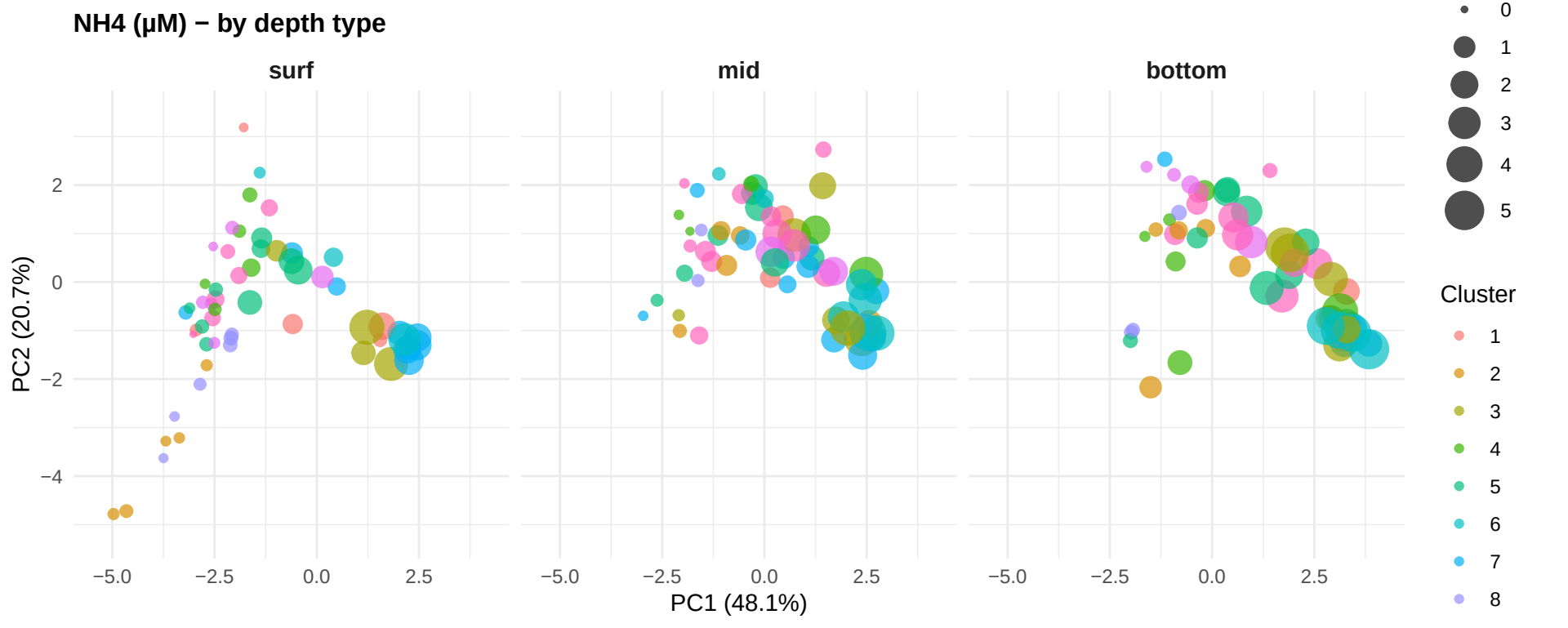

PO4 (µM) − all depths
PO4 (µM)
0.5
1.0
1.5
Cluster
1
2
3
4
5
6
7
8
9
10
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5

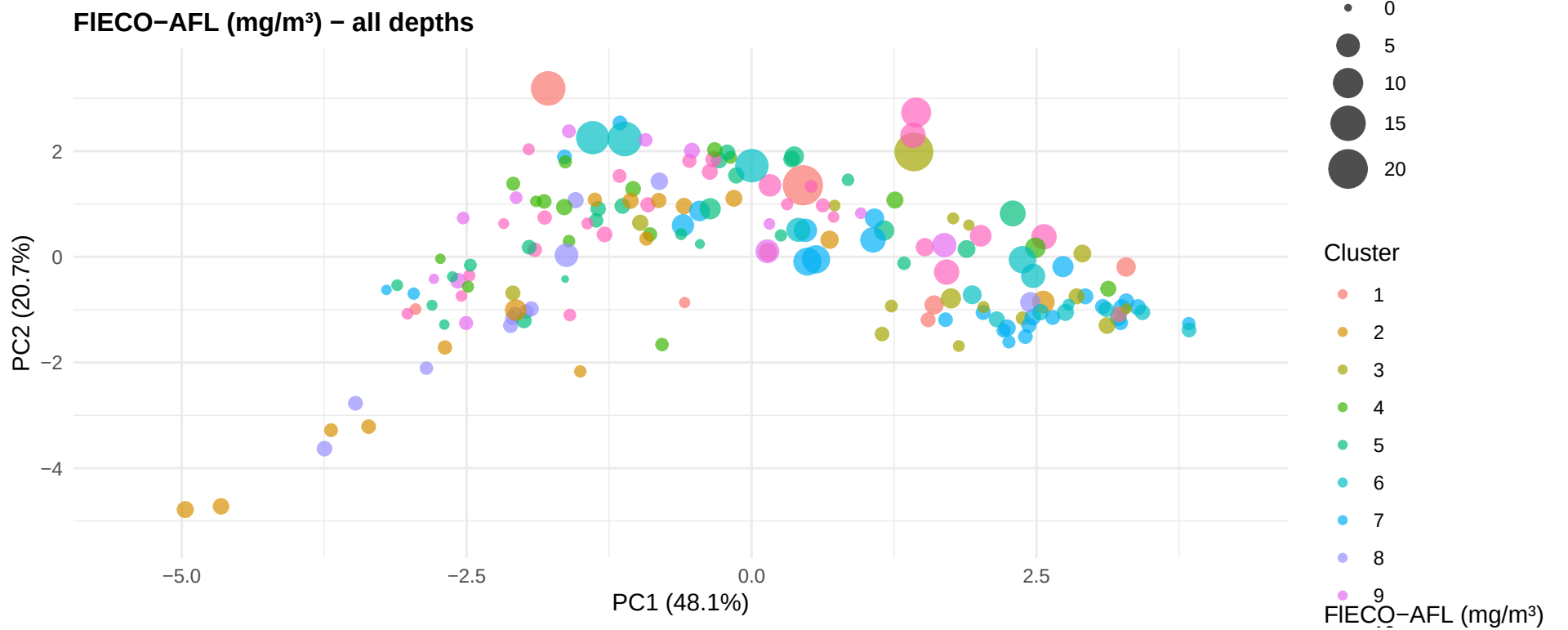

PO4 (µM) − by depth type
surf
mid
bottom
PO4 (µM)
0.5
1.0
1.5
Cluster
1
2
3
4
5
6
7
8
9
10
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5

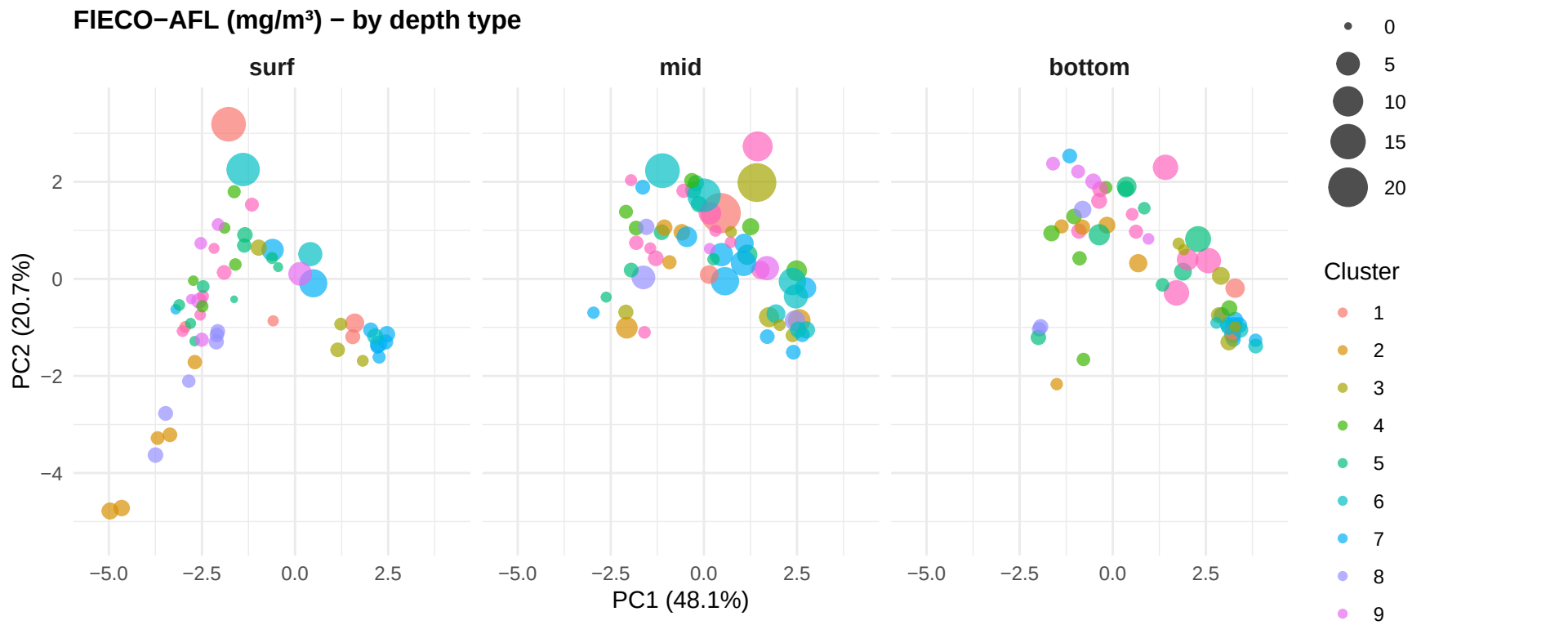

Silicate (µM) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Silicate (µM)
10
20
30
Cluster
1
2
3
4
5
6
7
8
9
10

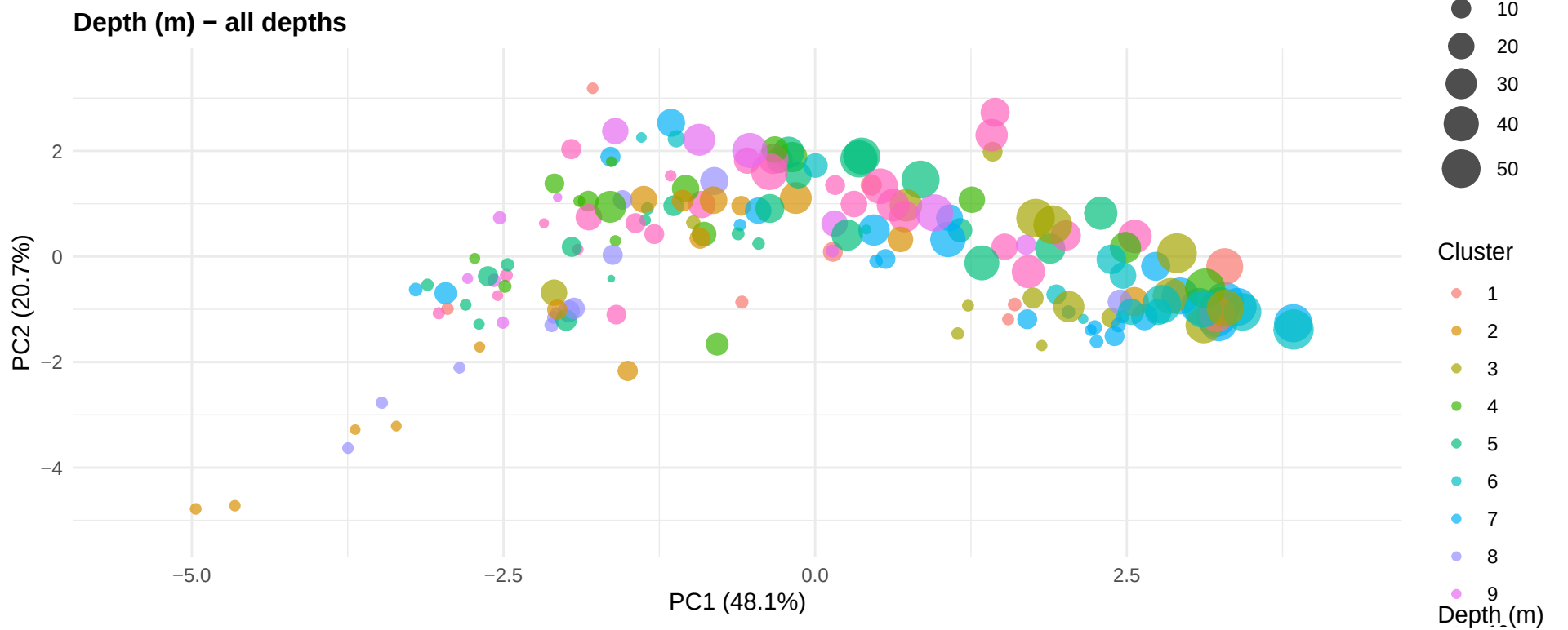

Silicate (µM) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Silicate (µM)
10
20
30
Cluster
1
2
3
4
5
6
7
8
9
10

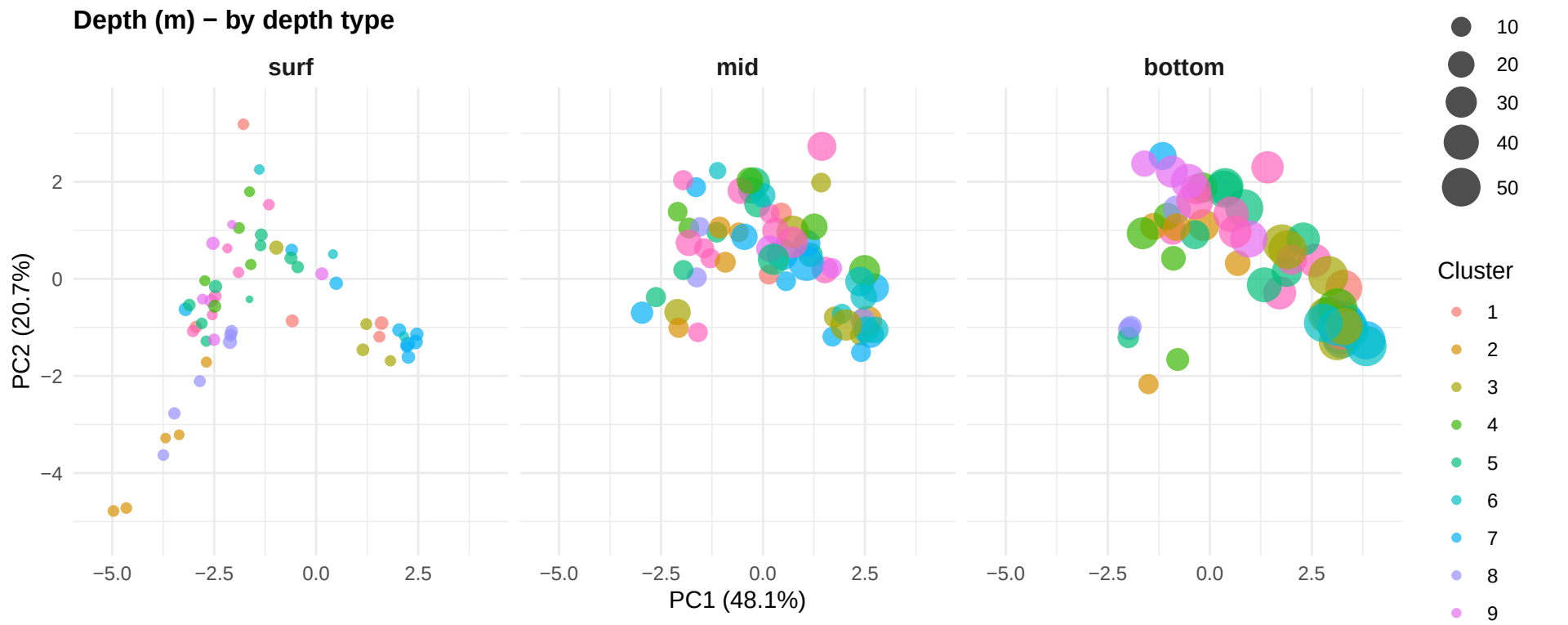

NH4 (µM) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
1
2
3
4
5
Cluster
1
2
3
4
5
6
7
8
9


NH4 (µM) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
NH4 (µM)
0
1
2
3
4
5
Cluster
1
2
3
4
5
6
7
8


FIECO−AFL (mg/m³) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
FIECO−AFL (mg/m³)
0
5
10
15
20
Cluster
1
2
3
4
5
6
7
8
9


FIECO−AFL (mg/m³) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
FIECO−AFL (mg/m³)
0
5
10
15
20
Cluster
1
2
3
4
5
6
7
8
9


Depth (m) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Depth (m)
10
20
30
40
50
Cluster
1
2
3
4
5
6
7
8
9


Depth (m) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Depth (m)
10
20
30
40
50
Cluster
1
2
3
4
5
6
7
8
9